1. **PERIODIC TABLE**

Development of PT
Revision quiz:

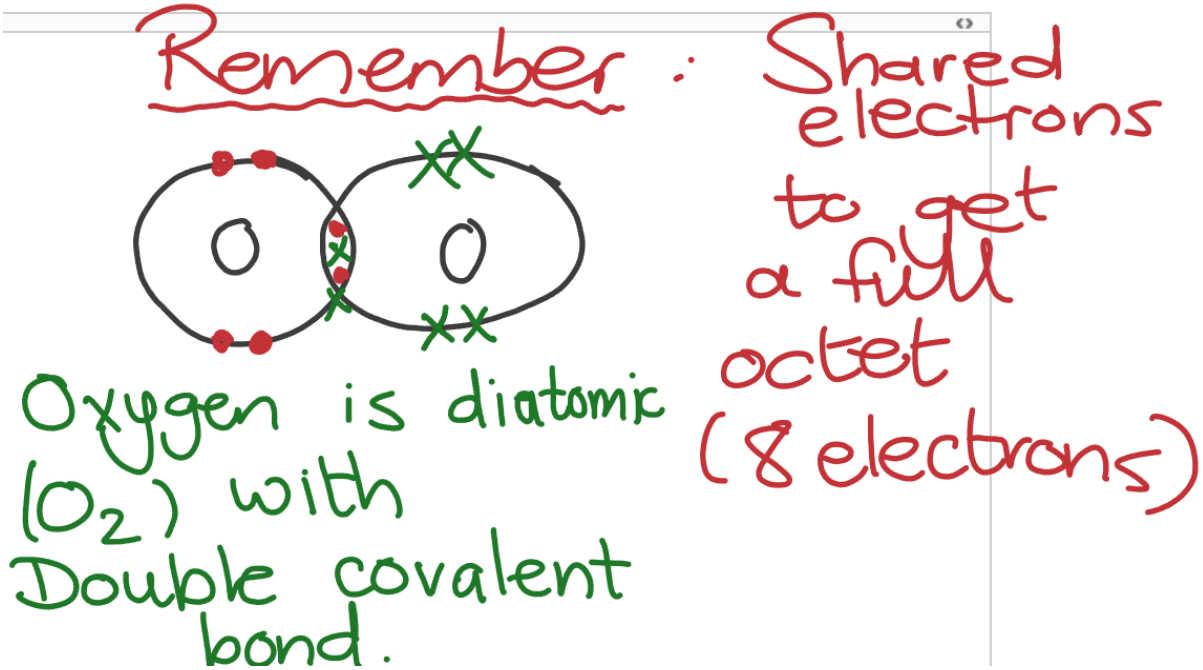
2. **History of the atom**:

Here are some videos to help you revise.
Atom Theory
Revision quiz

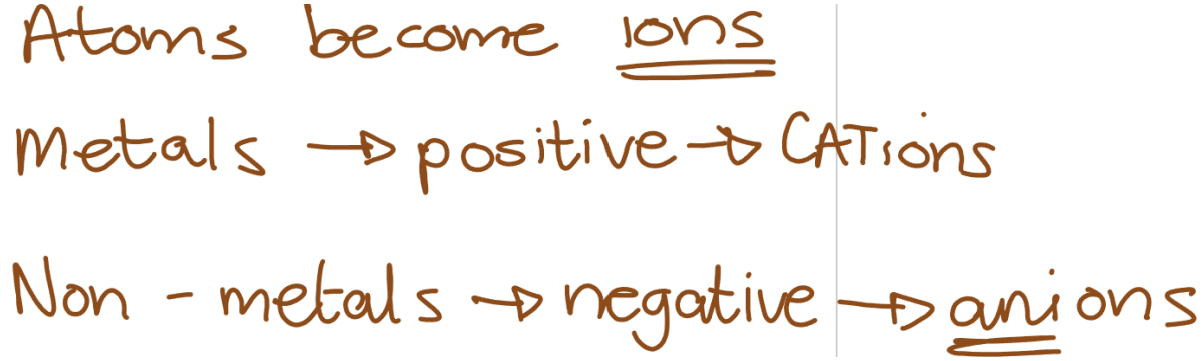
3. **Electronic configuration of the atom**

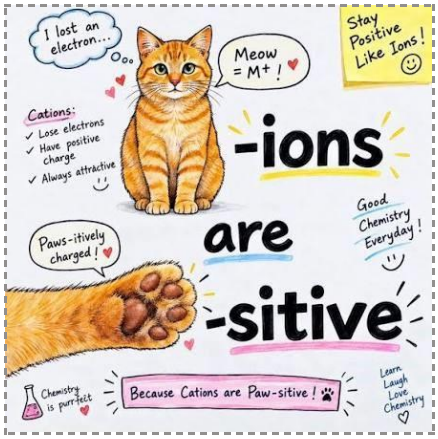
Structure of the Atom
Revision Quiz

1. Covalent Bonds - Non-Metal only



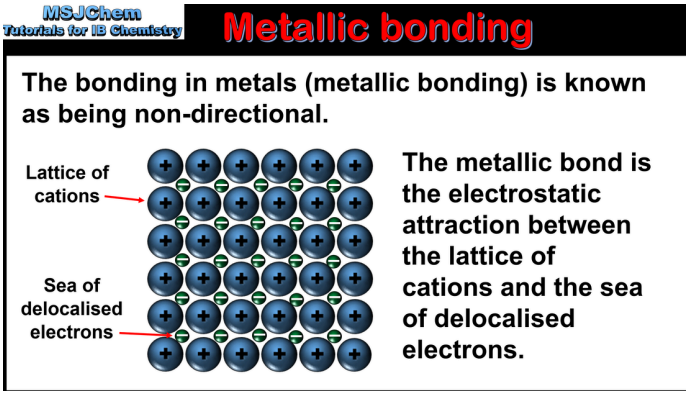
2. Ionic Bonds - Metal and Non-metal





3. Metallic Bonds - Metals Only

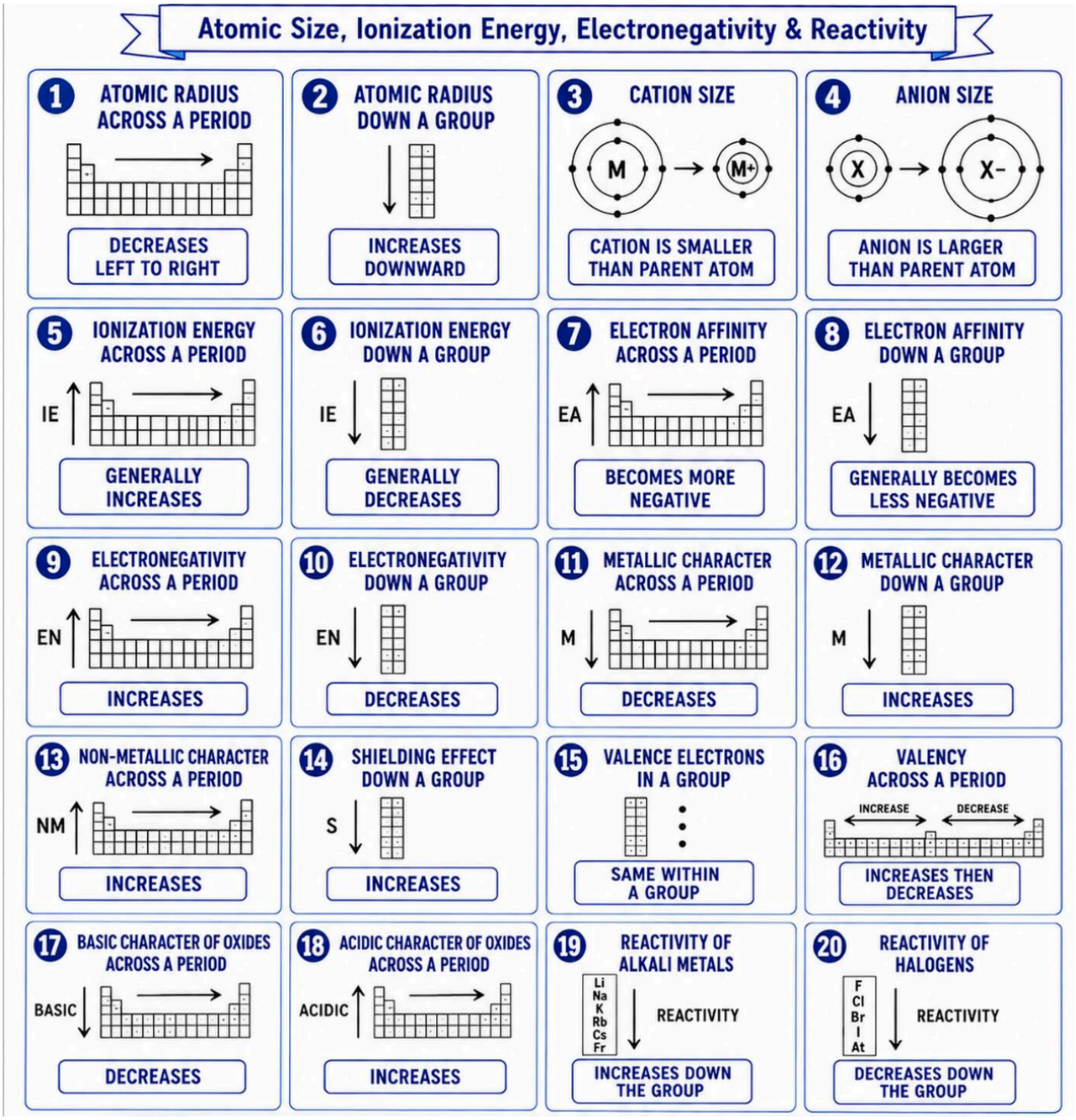
Electrons are a sea of delocalised electrons.

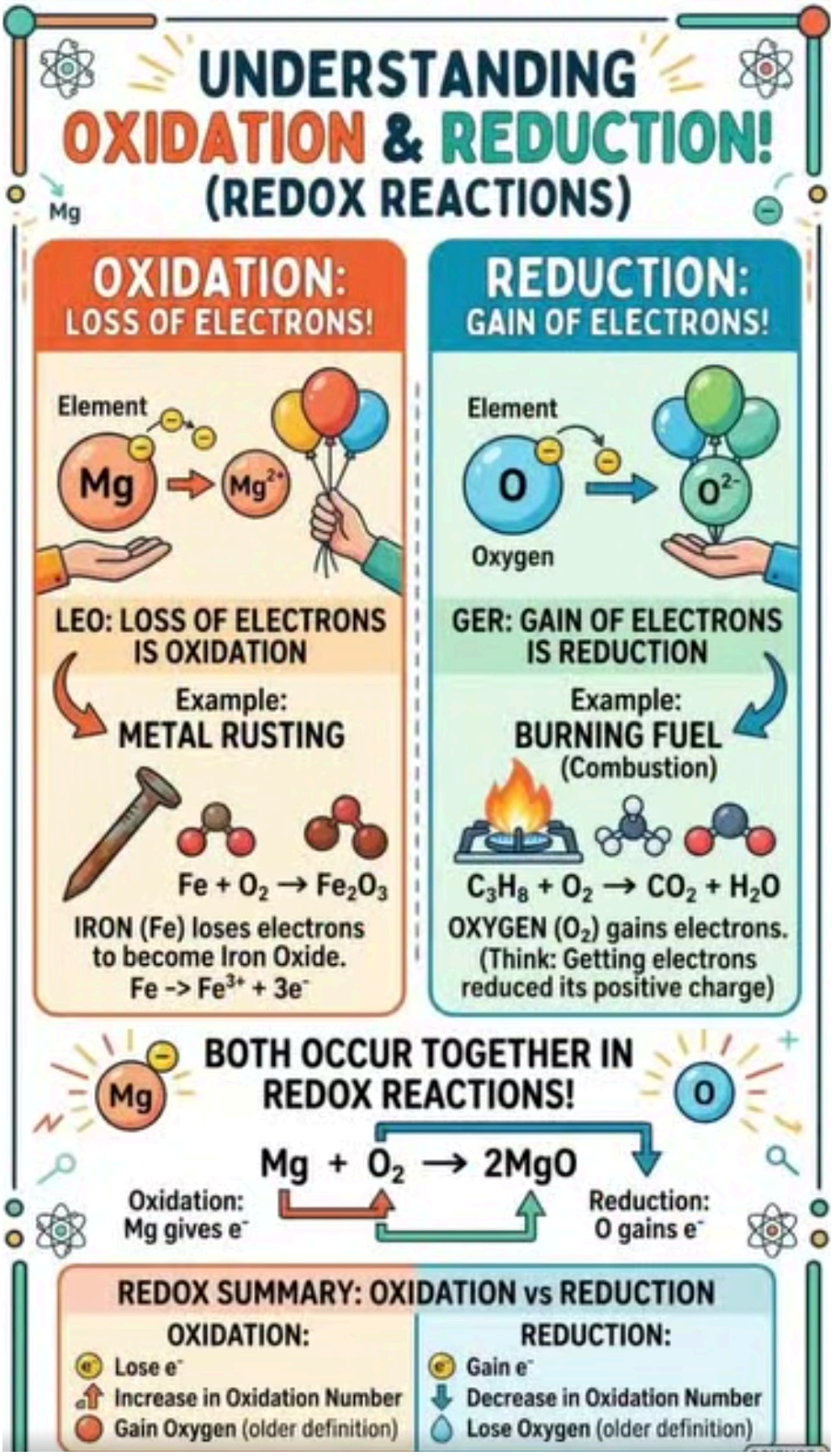


Summary

Chemical Notation (what do the numbers mean in chemistry formulae?)

[What is the reactivity series?]





REdox reaction

Testing for Sugars (Glucose)

- Sugars are the building blocks for carbohydrates
- Glucose is used for respiration
- Foods that contain glucose include sweets and fruits

1. Prepare food sample
2. Add Benedict's reagent
3. Heat with a Bunsen burner or water bath
4. Look at the colour





Blue has NO GLUCOSE



Brick Red contains GLUCOSE

Safety considerations

- Keep anything flammable away from Bunsen burner
- Careful to avoid burns

Testing for Lipids

- Lipids are fats and oils
- They are used for energy and insulation
- Foods that contain lipids include butter and oily fish

1. Prepare food sample
2. Add Ethanol & Water
3. Shake
4. Look at the colour





Colourless and Clear has NO LIPIDS



Cloudy White contains LIPIDS

Safety considerations

- Ethanol is highly flammable keep away from flames
- Ethanol is toxic, do not consume

Testing for Starch

- Starch is a complex carbohydrate found in plants
- It is how plants store glucose
- They are used for energy
- Foods that contain starch include bread and pasta

1. Prepare food sample
2. Add iodine
3. Look at the colour





Orange has NO STARCH



Blue-Black contains STARCH

Safety considerations

- Iodine is an irritant
- Wear goggles

Testing for Proteins

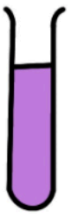
- Proteins are large molecules made of chains of amino acids
- They are used for growth and repair
- Foods that contain protein include meat and fish

1. Prepare food sample
2. Add Biuret reagent
3. Look at the colour





Blue has NO PROTEIN



Purple contains PROTEIN

Safety considerations

- Biuret is corrosive
- Wear goggles
- Clean up spill ASAP

BIOLOGY SAVER

LABORATORY GAS IDENTIFICATION TESTS

GAS	IDENTIFICATION TEST	OBSERVATION
H_2 Hydrogen (H_2)	Bring a burning splint near the gas	Burns with a 'pop' sound
O_2 Oxygen (O_2)	Insert a glowing splint	Glowing splint relights
CO_2 Carbon Dioxide (CO_2)	Bubble through limewater	Limewater turns milky
NH_3 Ammonia (NH_3)	Hold moist red litmus paper	Turns blue
Cl_2 Chlorine (Cl_2)	Hold moist blue litmus paper	Turns red, then bleaches white
SO_2 Sulfur Dioxide (SO_2)	Acidified $KMnO_4$ solution	Purple solution becomes colorless
H_2S Hydrogen Sulfide (H_2S)	Lead acetate paper	Turns black (PbS formed)

KEY REACTIONS

$CO_2 + Ca(OH)_2 \rightarrow CaCO_3\downarrow + H_2O$

$H_2S + Pb(CH_3COO)_2 \rightarrow PbS\downarrow + 2CH_3COOH$

$SO_2 + KMnO_4$ (acidic) $\rightarrow$ Mn^{2+} (colorless) + products

QUICK MEMORY TRICKS

- H_2 → "Pop"
- O_2 → Relights glowing splint
- CO_2 → Limewater milky
- NH_3 → Red litmus → Blue
- Cl_2 → Blue litmus → Red → White
- SO_2 → Decolorizes acidified $KMnO_4$
- H_2S → Black lead acetate paper

EXAM TIP

Always remember the **test reagent** and the **characteristic observation** together—they are frequently asked in practical exams and competitive chemistry tests.

CHEMISTRY — PREPARATION —

PREPARE • PRACTICE • SUCCEED